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Cancer Center

Introduction to Cancer Genomics & MSK-IMPACT

GEMS Training

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Agenda

- Genomics
 - Cancer genomics
 - Precision oncology
 - Genetic mutations
 - Genomic sequencing
- MSK–IMPACT
 - MSK genomics
 - IMPACT data – {cbioportalR}
 - IMPACT analyses and visualizations – {gnomeR}



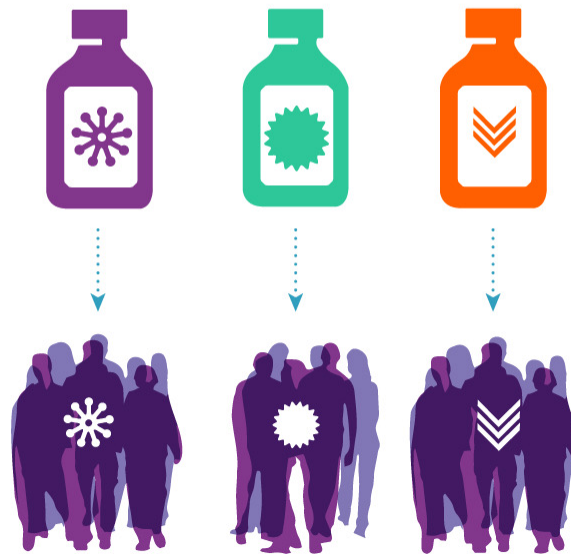
Cancer is a Genetic Disease



- Cancer is a collection of diseases characterized by uncontrolled cell growth
- Cancer cells can divide rapidly, evade protective mechanisms that kill abnormal cells, and metastasize
- These changes are caused by a series of somatic mutations

UNDERSTANDING PRECISION MEDICINE

In precision medicine, patients with tumors that share the same genetic change receive the drug that targets that change, no matter the type of cancer.



Credit: National Cancer Institute



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What is Precision Oncology?



National Human Genome
Research Institute

Est 1989

"An emerging medical discipline that involves using genomic information about an individual as part of their clinical care (e.g., for diagnostic or therapeutic decision-making) and the health outcomes and policy implications of that clinical use."



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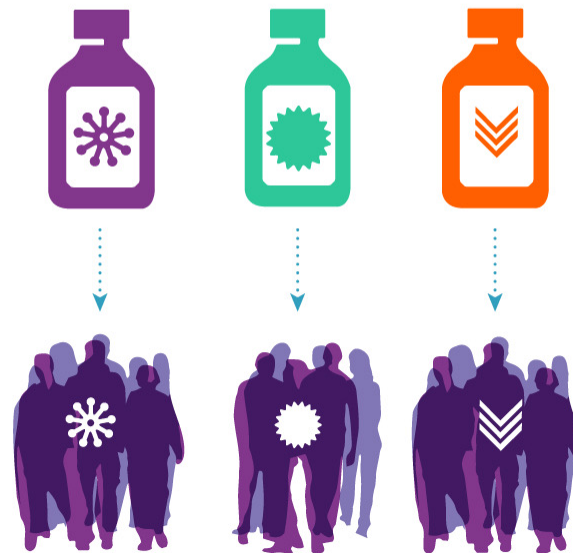
What is Precision Oncology?



*Genomic information about cancer is leading to better diagnoses and treatment strategies that are tailored to patients' tumors, an approach called **precision medicine**.*

UNDERSTANDING PRECISION MEDICINE

In precision medicine, patients with tumors that share the same genetic change receive the drug that targets that change, no matter the type of cancer.



Credit: National Cancer Institute



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Cancer genomics success stories – targeted therapies

HER2+
Breast Cancer
“worst to first”



Trastuzumab (Herceptin) controls the hyper-active signaling pathway (HER2 tyrosine kinase) caused by multiple copies of the *HER2* gene on chr 17 in a subtype of breast cancers. (~1998)

BCR-ABL+
Myeloid Leukemia



Chromosomal rearrangement creates oncogene BCR-ABL in Chronic Myeloid Leukemia. Imatinib (Gleevec) inhibits overactivity of the Bcr-Abl tyrosine kinase protein. (~2001)

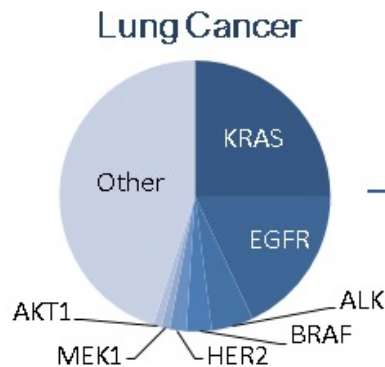
EGFR+
Non Small Cell
Lung Cancer



Erlotinib restricts activation of a protein (epidermal growth factor, or EGFR), which is mutated and abnormally active in a subset of lung cancers. (~2004)

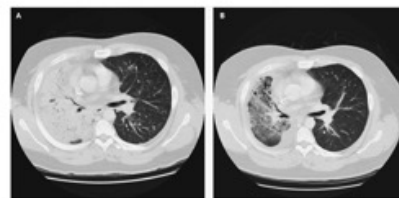
Cancer genomics success stories – targeted therapies

Precision Oncology

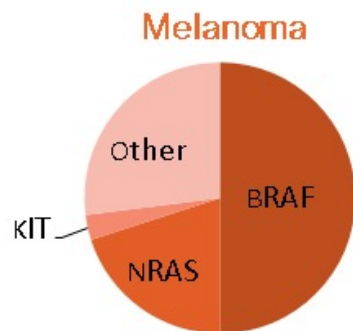


EGFR-mutant cancer

EGFR inhibitors:
erlotinib, gefitinib

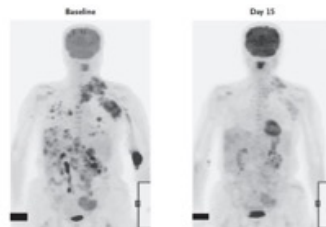


Lynch et al., NEJM, 2004
Paez et al., Science, 2004
Pao et al., PNAS, 2004



BRAF-mutant cancer

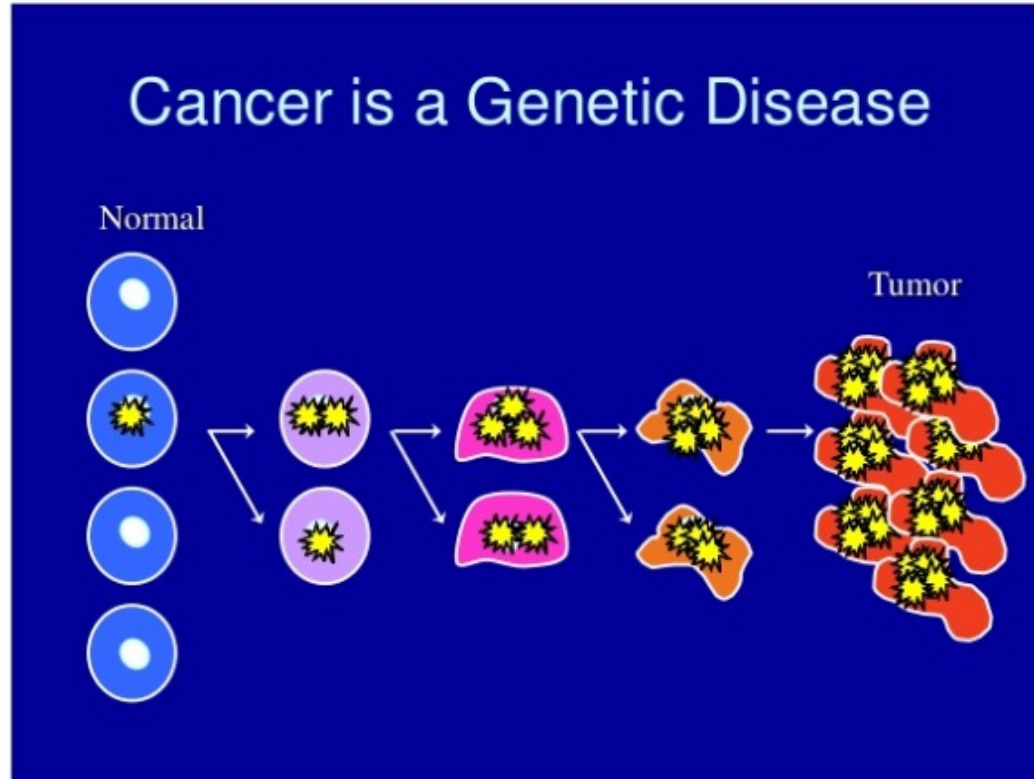
BRAF inhibitor:
vemurafenib, dabrafenib



Flaherty et al., NEJM, 2010

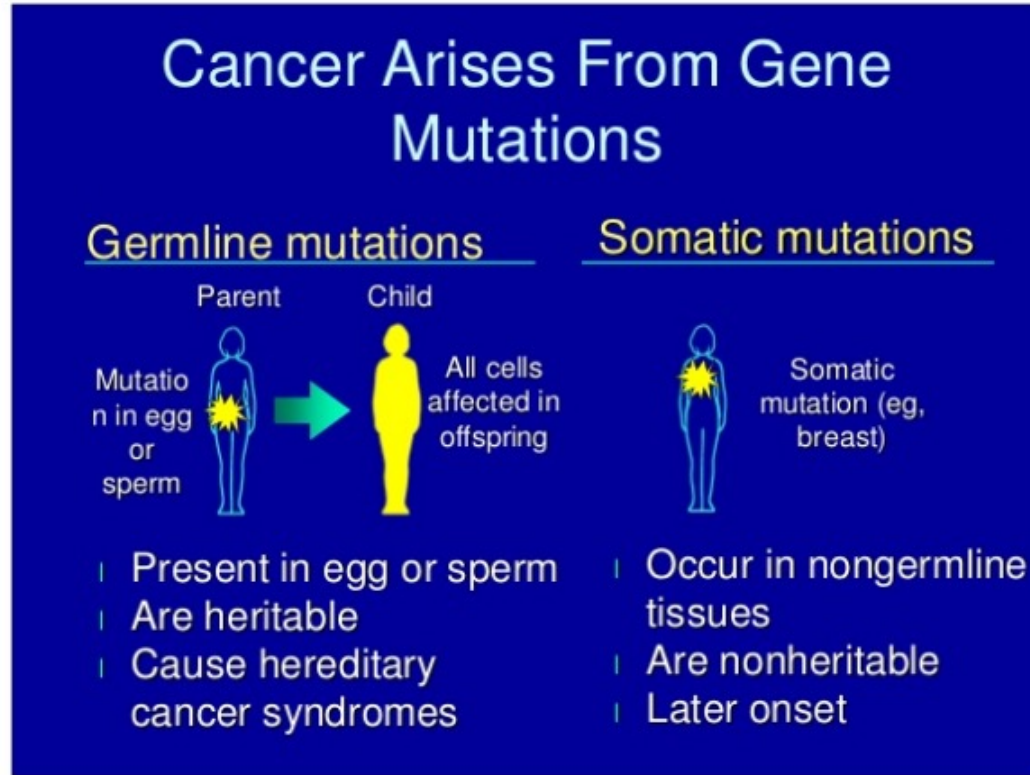


Cancer genomics – genetics review



The transition from a normal cell to malignant cancer is driven by DNA mutations and alterations.

Cancer genomics - genetics

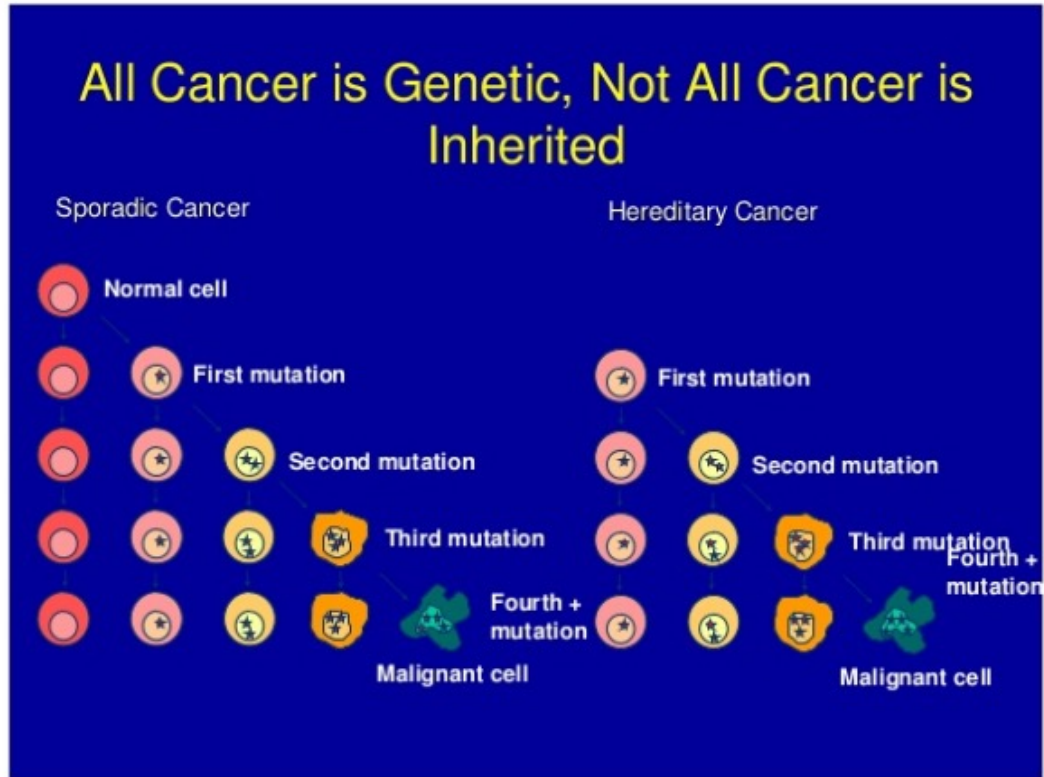


From Hereditary Cancer Predisposition, Slides by Patricia Ganz, MD, UCLA



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Cancer genomics - genetics



From Hereditary Cancer Predisposition, Slides by Patricia Ganz, MD, UCLA

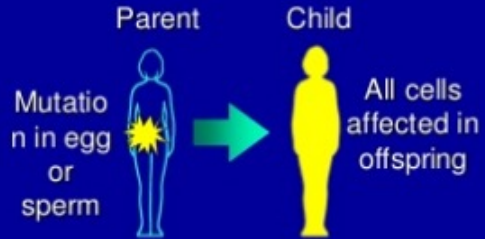


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Cancer genomics – genetics

Cancer Arises From Gene Mutations

Germline mutations



- | Present in egg or sperm
- | Are heritable
- | Cause hereditary cancer syndromes

Somatic mutations



- | Occur in nongermline tissues
- | Are nonheritable
- | Later onset

5 -10% of cancers

90 - 95% of cancers

From Hereditary Cancer Predisposition, Slides by Patricia Ganz, MD, UCLA



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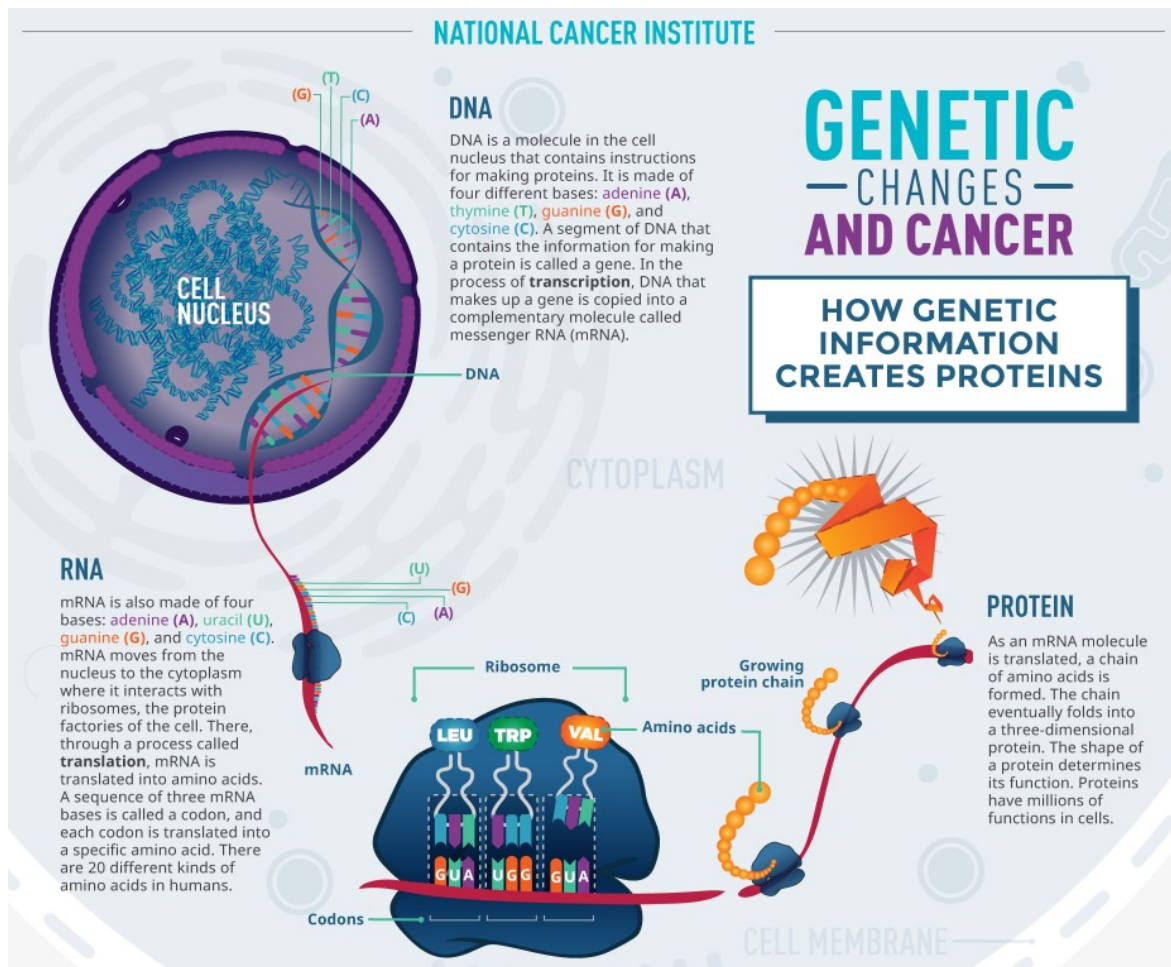
Cancer genomics – somatic mutations

Possible causes:

- Spontaneous occurrences at low frequency
- Errors that occur as cells divide
- Exposure to carcinogenic substances, e.g.
 - Radiation (UV rays from sun)
 - Chemicals in tobacco smoke



Cancer genomics – somatic mutations



How a mutation causes damage:

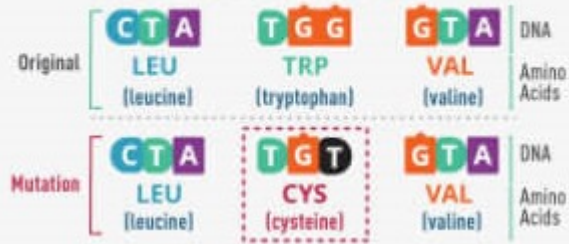
DNA to RNA to Protein



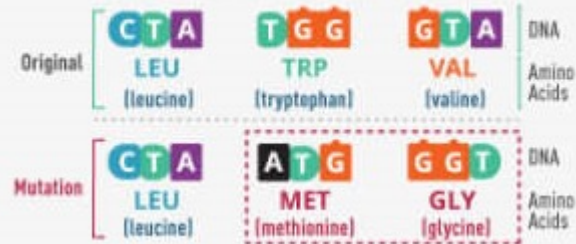
Cancer genomics – somatic mutations

TYPES OF GENETIC MUTATIONS IN CANCER

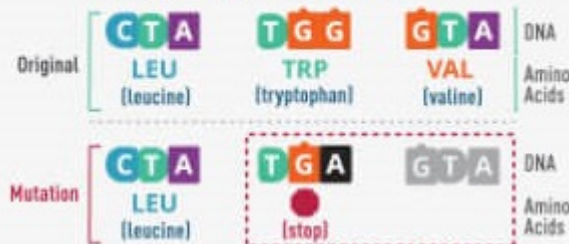
MISSENSE MUTATION



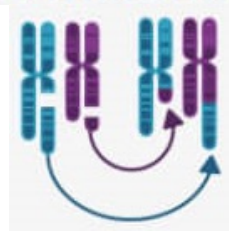
FRAMESHIFT MUTATION



NONSENSE MUTATION



CHROMOSOME REARRANGEMENTS



Mutations can change proteins, which can change cells behavior...



Cancer genomics – detecting somatic mutations

Tumor DNA sequencing assays

- Tumor sample from biopsy or resection
- Compares DNA sequence in cancer cells with that in normal cells
- Broad or targeted focus
 - Broad regions: whole genome / whole exome sequencing
 - Targeted regions: gene panels, e.g. MSK-IMPACT





MSK IMPACT

Integrated Mutation Profiling of Actionable Cancer Targets

Goal: Expanding genomic testing to more patients

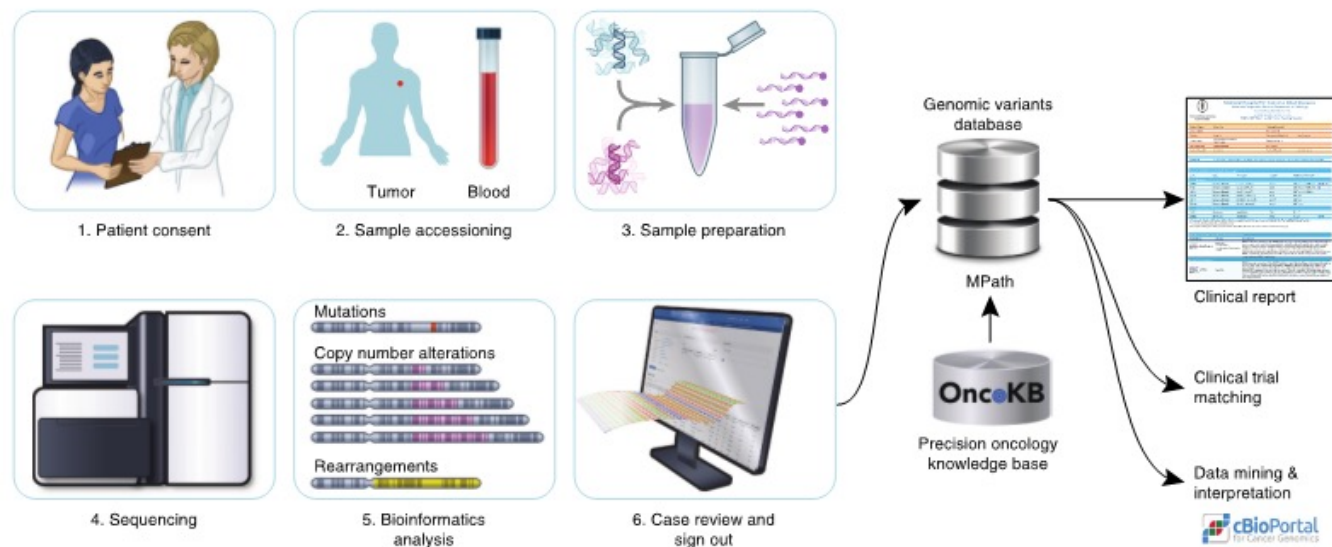
- Now routinely used in clinical practice to sequence patient tumors.
- Nov 2017: MSK-IMPACT was first tumor-profiling test to be authorized by US FDA



MSK IMPACT

Precision Oncology with MSK-IMPACT

Identifies **mutations**, **copy number alterations**, **structural variants** in **468 genes** as well as biomarkers that could predict treatment response such as **MSI status** or **tumor mutation burden (TMB)**



MSK IMPACT

MSK-IMPACT Genes

Panel includes exons, introns, and other noncoding regions

22,118 cases

341 genes: 2,861 cases

410 genes: 10,004 cases

468 genes: 9,253 cases

Cancer Gene Exons:

6,614 protein-coding exons of 468 genes

- actionable mutations
- targets of investigational agents
- frequently mutated in cancer
- cancer susceptibility genes

Cancer Gene Introns:

70 introns of 20 recurrently rearranged genes

Cancer Gene Promoters: TERT promoter

SNP Probes: >1,000 non-coding SNPs

- copy number analysis, various QC checks

Target Territory = 1.52 Mb

MSK IMPACT

MSK-IMPACT Genes

Panel includes exons, introns, and other noncoding regions

22,118 cases

341 genes: 2,861 cases

410 genes: 10,004 cases

468 genes: 9,253 cases



Updated as of 7/22:

~80,652 cases:

468 genes: 36,636 cases

505 genes: 17,324 cases

Cancer Gene Exons:

6,614 protein-coding exons of 468 genes

- actionable mutations
- targets of investigational agents
- frequently mutated in cancer
- cancer susceptibility genes

Cancer Gene Introns:

70 introns of 20 recurrently rearranged genes

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MSK IMPACT - OncoKB

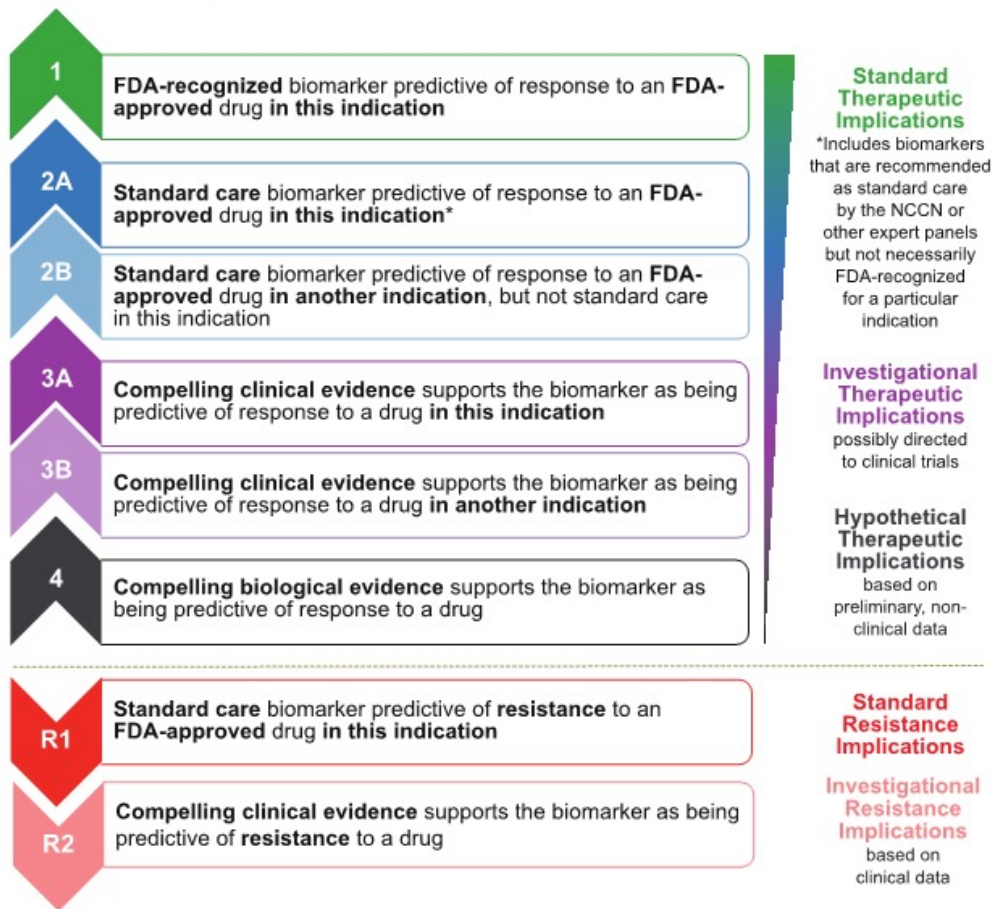
Goal:

Help doctors find best therapies based on tumor's genomic profile

OncoKB – Oncology Knowledge Base

Contains info about likely oncogenicity and clinical actionability of specific cancer gene alterations

OncoKB Therapeutic Levels of Evidence

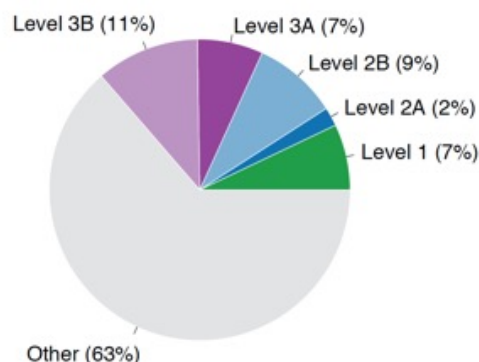


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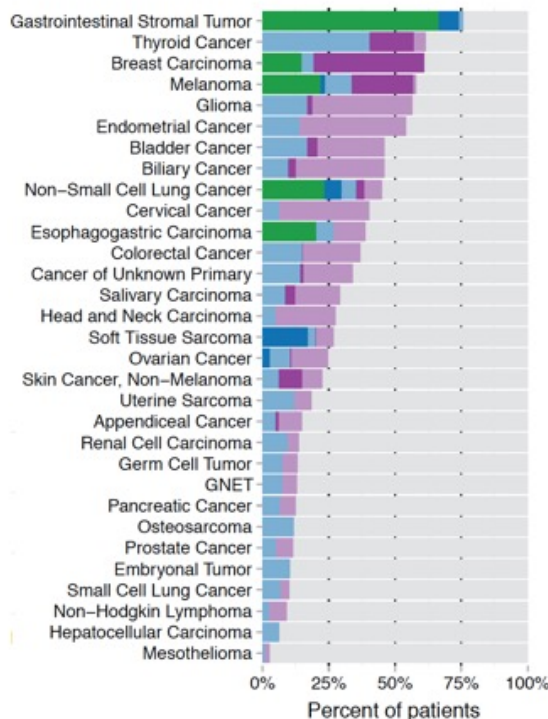
MSK IMPACT - OncoKB

Prevalence of Actionable Mutations

37% of patients harbored ≥ 1 OncoKB-annotated actionable mutation



Level 1	FDA-recognized biomarker for an FDA-approved drug in the same indication
Level 2A	Standard of care biomarker for an FDA-approved drug in the same indication
Level 2B	Standard of care biomarker for an FDA-approved drug in another indication
Level 3A	Compelling clinical evidence supporting the biomarker as being predictive of drug response in the same indication
Level 3B	Compelling clinical evidence supporting the biomarker as being predictive of drug response in another indication



MSK IMPACT – Data sharing

Goal:

Gather data and share analyses within MSK and with the public.



- MSK-IMPACT results stripped of patient-identifying info available for scientific use by everyone at MSK through [cBioPortal](#) and by greater scientific community through AACR.
- cBioPortal provides **visualization, analysis** and **download** of published data sets.
- As of the end of 2018, more than 100 scientific publications by MSK authors have included results from MSK-IMPACT testing.



MSK IMPACT – Data sharing

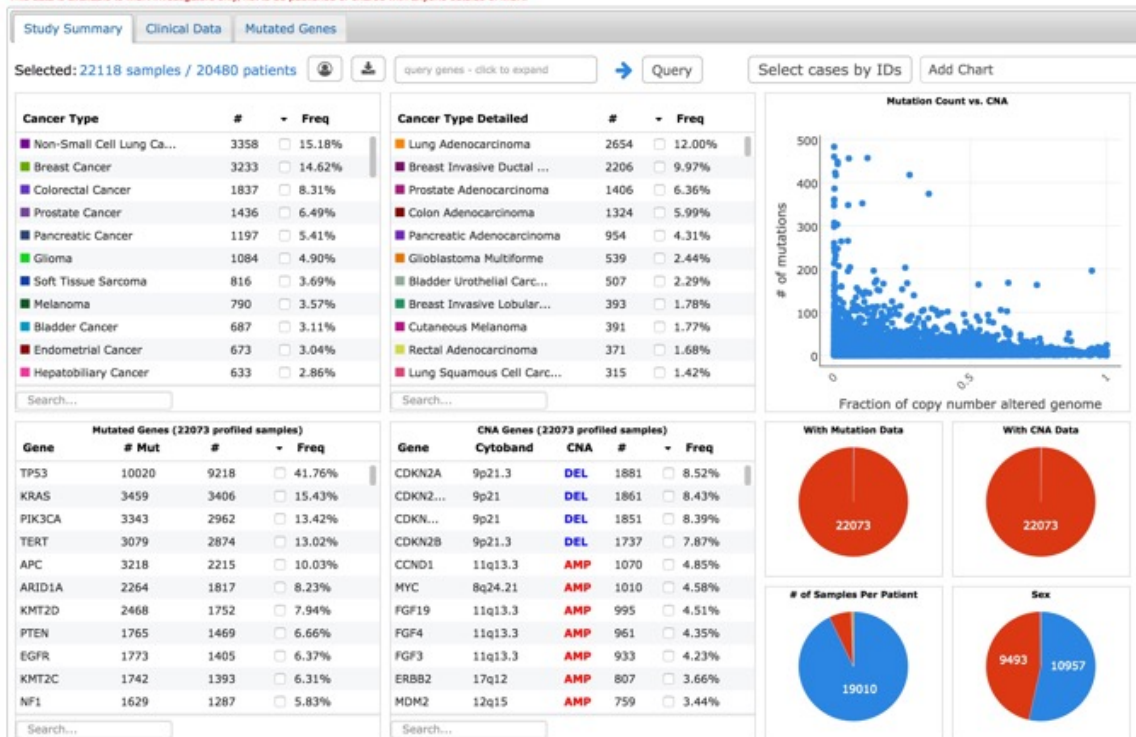
Data sharing with Researchers



Cohort level data

MSK-IMPACT Clinical Sequencing Cohort (MSKCC) [Query this study](#)

Targeted sequencing of clinical cases via MSK-IMPACT. Please follow the [publication guidelines](#) when using this data in abstracts or journal articles. This data is available to MSK investigators only, not to be published or shared with anyone outside of MSK.



MSK IMPACT – Data for Analysis



- Data from studies in progress are also deposited in cBioPortal and investigators receive specific access to these studies
- Questions of interest:
 - Genes / mutations / pathways associated with prognosis / prediction
 - Genes / mutations / pathways associated with clinical characteristics, e.g. Stage, grade, lymphovascular invasion, perineural invasion, etc
 - Is there co-occurrence or mutual exclusivity among genes in the cohort?
 - Are there clinically meaningful subtypes in the cohort?
- Approaches to obtaining IMPACT data:
 - Ask for copy of **annotated** cBioPortal data
 - Essential if you want to look at location of mutation or **oncogenicity**, among other characteristics.
 - Download data yourself with **cbioportalR** package



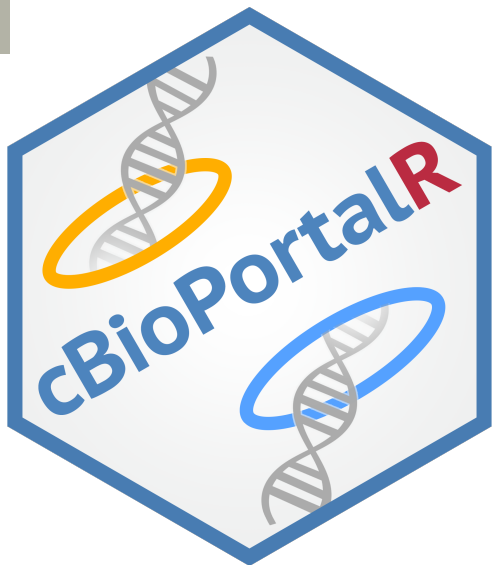
MSK IMPACT – Data for Analysis



- Obtaining IMPACT data:
 - Download data yourself with **cbioportalR** package
 - Either with **sample ids** or **study id**



MSK IMPACT – cbioportalR



{cbioportalR} allows you to access [cBioPortal's](#) genomic and clinical data sets directly through R

- **Install the most up to date version:**
`remotes::install_github("karissawhiting/cbioportalR")`

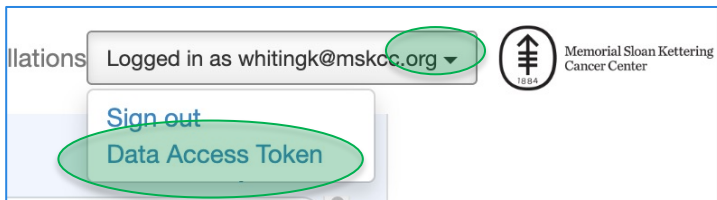
- **Documentation:**

C



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MSK IMPACT – cbioportalR

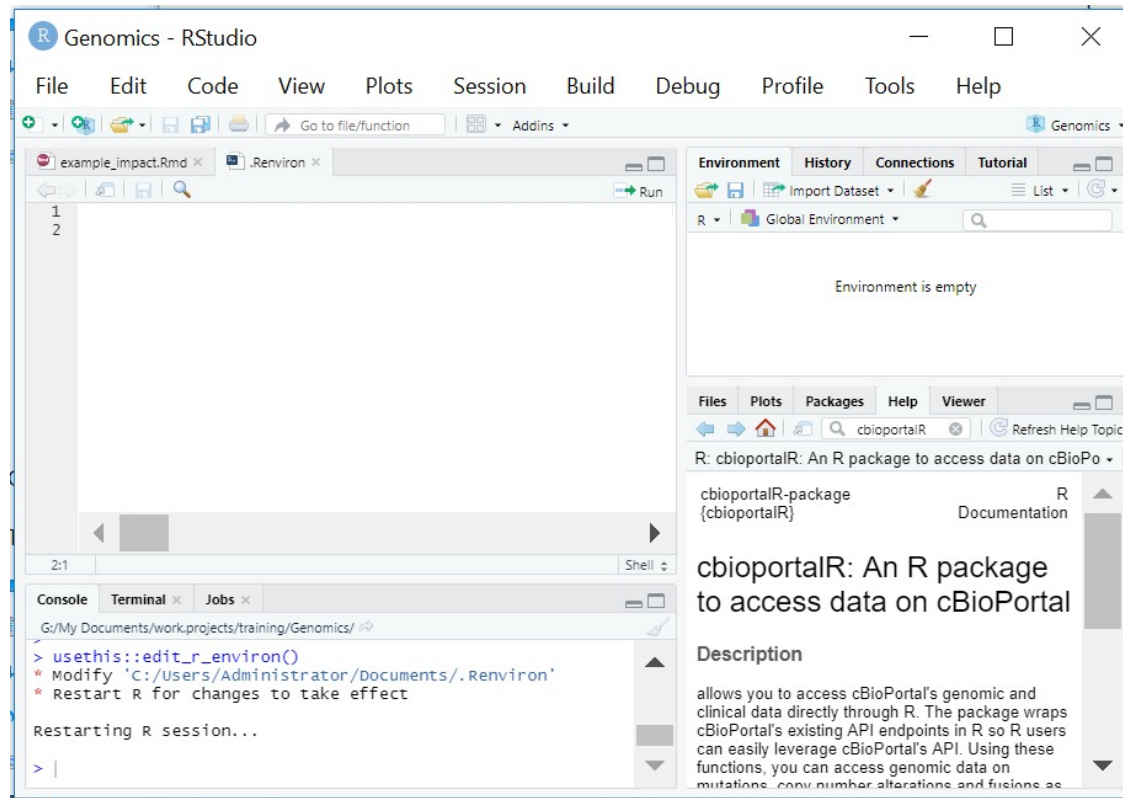


(more detailed instructions on website)

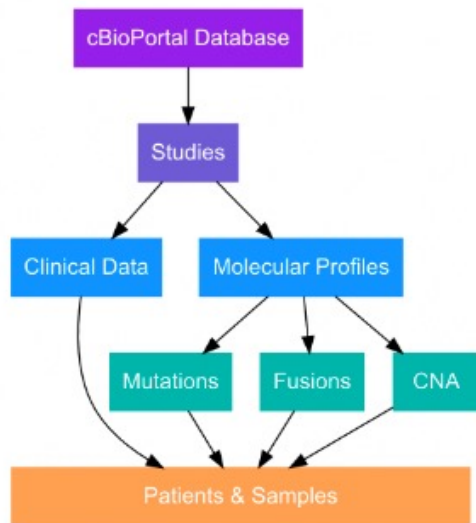
- You can access public data anytime without credentials:
 - `set_cbioportal_db("public")`
- For projects you will need to authenticate yourself to use internal MSK Data:
 - Log in and get a token from <https://cbioportal.mskcc.org>
 - Save it to your R environment
`usethis::edit_renv()`
-  Reload R, then: `set_cbioportal_db("msk")`



MSK IMPACT – cbioportalR



MSK IMPACT – How Data is Organized in cBioPortal

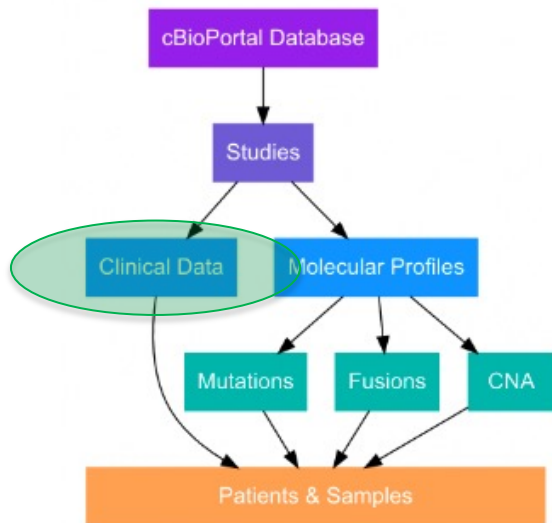


Level	Functions
Database	<code>`available_studies()`</code> , <code>`available_profiles()`</code> , <code>`available_gene_panels()`</code> , <code>`get_genes()`</code>
Studies	<code>`available_profiles()`</code> , <code>`get_study_info()`</code> , <code>`available_clinical_attributes()`</code> , <code>`get_genetics_by_study()`</code> , <code>`get_mutations_by_study()`</code> , <code>`get_cna_by_study()`</code> , <code>`get_fusions_by_study()`</code> , <code>`get_clinical_by_study()`</code> , <code>`available_samples()`</code> , <code>`available_patients()`</code>
Molecular Profiles	<code>`get_genetics_by_study()`</code> , <code>`get_mutations_by_study()`</code> , <code>`get_cna_by_study()`</code> , <code>`get_fusions_by_study()`</code>
Patients & Samples	<code>`get_genetics_by_sample()`</code> , <code>`get_mutations_by_study()`</code> , <code>`get_cna_by_sample()`</code> , <code>`get_fusions_by_sample()`</code> , <code>`get_clinical_by_sample()`</code> , <code>`get_clinical_by_patient()`</code> , <code>`get_panel_by_sample()`</code> , <code>`get_samples_by_patient()`</code>



MSK IMPACT – How Data is Organized in cBioPortal

You can also pull other sample-level or patient-level characteristics like: age, sex, TMB, cancer type, etc.



Level	Functions
Database	<code>`available_studies()`</code> , <code>`available_profiles()`</code> , <code>`available_gene_panels()`</code> , <code>`get_genes()`</code>
Studies	<code>`available_profiles()`</code> , <code>`get_study_info()`</code> , <code>`available_clinical_attributes()`</code> , <code>`get_genetics_by_study()`</code> , <code>`get_mutations_by_study()`</code> , <code>`get_cna_by_study()`</code> , <code>`get_fusions_by_study()`</code> , <code>`get_clinical_by_study()`</code> , <code>`available_samples()`</code> , <code>`available_patients()`</code>
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MSK IMPACT – data niggling details



IMPACT ID format:

- P-0000113-T01-IM3
- Patient ID-Tumor sample number-IMPACT panel version
- Samples with other prefixes were not sequenced by cBioPortal

Number of samples in study may not correspond to number of samples in genetic data

- Some samples may not have any mutations or copy number alteration

Data formats may vary

- Differences in mut vs cna data - Tumor_sample_barcode vs SampleId
- CNA often in wide format and may need transformation

Mutation_Status

- Categories: GERMLINE, SOMATIC, UNKNOWN. In many cases, UNKNOWN are of interest. Be sure to filter out germline unless you are specifically studying these.

Impact panels

- IM3 - 341 genes
- IM5 - 410 genes
- IM6 – 468 genes
- IM7 – 505 genes

Example scripts will be available:

<https://github.com/MSKCC-Epi-Bio/introduction-to-msk-impact>



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MSK IMPACT – gnomeR

- Documentation: <https://mskcc-epi-bio.github.io/gnomeR/>
- gnomeR Slides: https://genie-bpc.github.io/intro_to_genieBPC_and_gnomeR/geniebpc-gnomer-r-med.pdf





Introduction to Genomics & MSK-IMPACT data

Thanks for
your attention!



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